Practice Quiz 10B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 10

Strong induction for a representation by nonnegative integers.

1. Prove by strong induction that every integer $n \geq 4$ can be written

$$n = 2a + 5b$$

for some nonnegative integers a, b. State the complete inductive hypothesis and verify every required base case.

Solution We use strong induction. Let $n \geq 4$ and assume that for every integer k with $4 \leq k < n$, there are nonnegative integers a, b such that $k = 2a + 5b$.

For $n = 4$, take $a = 2, b = 0$, so $4 = 2(2) + 5(0)$. For $n = 5$, take $a = 0, b = 1$, so $5 = 2(0) + 5(1)$.

Now suppose $n \geq 6$. Then $4 \leq n - 2 < n$, so the inductive hypothesis gives nonnegative integers a, b with $n - 2 = 2a + 5b$. Therefore

$$n = 2a + 5b + 2 = 2(a + 1) + 5b.$$

The integers $a + 1$ and b are nonnegative, so this is the required representation of n. Strong induction proves the claim for every integer $n \geq 4$. ∎

Solutions revised using GPT-6 Astra in the Codex harness.